

DURANT, OK, USA

WMO: 742513

Lat: 33.942N

Lon: 96.394W

Elev: 698

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.6	25.1	9.4	9.1	27.6	13.7	11.3	33.0	26.0	49.4	24.0	51.9	8.1	340	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	100.4	76.1	99.0	75.8	96.7	75.6	80.0	92.2	78.7	91.2	77.9	90.6	8.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	144.2	85.4	75.3	135.9	83.5	74.7	133.2	83.0	44.1	91.3	42.8	90.9	41.7	91.2	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.3	20.2	18.3	13.6	103.7	3.5	3.7	11.1	106.4	9.1	108.6	7.1	110.7	4.6	113.4		
			12.1	81.6	3.0	1.4	10.0	82.6	8.2	83.4	6.5	84.2	4.4	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.7	44.0	46.7	56.3	63.4	71.9	80.6	83.8	83.8	77.0	66.0	55.3	46.0
	DBStd	16.22	9.74	10.57	9.14	7.87	6.71	4.36	4.43	4.88	5.84	7.97	8.73	9.06
	HDD50	692	236	171	44	7	0	0	0	0	0	2	45	187
	HDD65	2586	651	516	291	119	21	0	0	0	1	87	308	592
	CDD50	6042	51	79	240	408	678	918	1048	1046	810	499	203	62
	CDD65	2459	1	4	22	70	234	468	583	581	360	118	16	2
	CDH74	27107	7	30	166	581	2090	5134	7131	7098	3627	1114	117	12
	CDH80	13460	0	4	19	109	736	2548	3957	3976	1732	363	16	0
Wind	WSAvg	8.0	8.5	9.1	9.8	9.9	8.8	7.7	6.4	6.1	6.3	7.2	8.4	8.2
Precipitation	PrecAvg	43.9	2.1	2.8	3.9	4.2	5.4	5.0	2.8	2.9	5.1	4.2	3.3	2.8
	PrecMax	60.0	6.8	7.2	9.8	12.7	12.6	12.5	8.7	9.4	11.5	17.8	10.0	8.2
	PrecMin	21.3	0.0	0.1	1.0	0.3	0.6	1.6	0.0	0.0	0.2	0.2	0.0	0.3
	PrecStd	9.5	1.6	1.6	2.2	3.0	3.2	3.0	2.3	2.1	3.3	3.4	2.3	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	78.8	82.0	87.8	92.9	99.7	104.2	106.1	99.9	90.8	81.8	75.1
		MCWB	58.9	62.5	65.3	69.5	73.6	77.2	76.4	76.0	73.1	71.3	65.6	62.0
	2%	DB	70.2	72.5	78.9	82.2	90.3	97.0	100.3	101.8	96.8	88.1	77.3	70.2
		MCWB	58.2	60.3	64.3	67.6	73.6	75.2	76.2	76.2	73.3	71.1	63.8	59.1
	5%	DB	64.3	68.3	74.7	80.8	87.9	94.7	98.9	99.3	92.7	83.9	72.9	65.7
		MCWB	55.9	59.0	62.9	66.9	72.5	75.1	76.0	76.2	73.4	68.0	62.3	58.0
	10%	DB	61.2	63.9	71.9	77.3	83.9	91.1	96.5	96.7	90.1	81.0	70.2	62.6
		MCWB	51.8	56.5	61.0	65.1	71.9	75.7	76.2	75.9	72.8	66.9	61.4	55.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.9	66.4	69.3	73.4	78.2	80.8	82.1	81.8	78.8	75.8	69.5	66.7
		MCDB	70.0	73.4	76.9	81.8	88.1	92.7	93.8	94.0	89.8	83.7	74.6	72.5
	2%	WB	62.9	64.0	67.0	71.5	76.5	79.0	79.9	79.8	77.2	74.0	67.4	63.4
		MCDB	65.6	68.3	74.7	79.6	85.7	91.2	92.4	92.2	88.1	82.8	73.8	66.9
	5%	WB	58.0	61.2	65.5	69.8	75.0	77.9	78.5	78.4	75.9	72.3	65.5	60.1
		MCDB	63.9	66.6	72.3	76.7	83.3	89.8	91.4	91.6	86.4	80.2	71.2	63.9
	10%	WB	53.0	57.7	63.4	67.8	73.5	77.0	77.7	77.5	74.7	70.1	63.0	55.3
		MCDB	59.4	63.7	69.1	74.8	81.5	88.3	90.8	91.4	84.7	76.4	67.9	59.9
Mean Daily Temperature Range	5% DB	MDBR	20.5	20.1	21.4	21.8	19.9	20.0	21.2	21.9	22.0	22.8	21.6	20.2
		MCDBR	27.3	26.2	24.9	23.9	22.8	23.1	25.1	25.2	25.5	26.0	24.8	26.5
		MCWBR	17.5	16.6	13.9	12.6	9.6	7.3	6.5	6.4	8.2	11.5	14.5	17.9
	5% WB	MCDBR	23.3	23.0	20.6	19.9	19.1	19.7	20.8	21.5	20.0	20.1	20.6	21.9
		MCWBR	18.0	16.9	13.0	13.2	9.9	8.2	7.5	7.3	8.0	10.9	15.6	18.5
Clear-Sky Solar Irradiance	taub	0.310	0.319	0.354	0.385	0.422	0.432	0.438	0.442	0.395	0.352	0.332	0.311	
	taud	2.491	2.483	2.405	2.333	2.295	2.304	2.325	2.295	2.375	2.483	2.480	2.492	
	Ebn at Noon	287	296	292	286	275	271	269	266	275	280	276	279	
	Edh at Noon	27	30	36	40	42	42	41	41	36	30	27	26	
All-Sky Solar Radiation	RadAvg	868	1027	1367	1667	1805	2049	2094	2094	1920	1603	1246	927	756
	RadStd	89	152	101	126	178	217	217	125	122	156	182	116	94

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page